

```
In [1]: !pip install numpy seaborn matplotlib pandas scikit-learn
```

Requirement already satisfied: numpy in /Users/thomas200905/Documents/Thomas/HCMUS/Third Year/Semester 8/Data Visualisation/.venv/lib/python3.12/site-packages (2.4.1)

Requirement already satisfied: seaborn in /Users/thomas200905/Documents/Thomas/HCMUS/Third Year/Semester 8/Data Visualisation/.venv/lib/python3.12/site-packages (0.13.2)

Requirement already satisfied: matplotlib in /Users/thomas200905/Documents/Thomas/HCMUS/Third Year/Semester 8/Data Visualisation/.venv/lib/python3.12/site-packages (3.10.8)

Requirement already satisfied: pandas in /Users/thomas200905/Documents/Thomas/HCMUS/Third Year/Semester 8/Data Visualisation/.venv/lib/python3.12/site-packages (2.3.3)

Requirement already satisfied: scikit-learn in /Users/thomas200905/Documents/Thomas/HCMUS/Third Year/Semester 8/Data Visualisation/.venv/lib/python3.12/site-packages (1.8.0)

Requirement already satisfied: contourpy>=1.0.1 in /Users/thomas200905/Documents/Thomas/HCMUS/Third Year/Semester 8/Data Visualisation/.venv/lib/python3.12/site-packages (from matplotlib) (1.3.3)

Requirement already satisfied: cycler>=0.10 in /Users/thomas200905/Documents/Thomas/HCMUS/Third Year/Semester 8/Data Visualisation/.venv/lib/python3.12/site-packages (from matplotlib) (0.12.1)

Requirement already satisfied: fonttools>=4.22.0 in /Users/thomas200905/Documents/Thomas/HCMUS/Third Year/Semester 8/Data Visualisation/.venv/lib/python3.12/site-packages (from matplotlib) (4.61.1)

Requirement already satisfied: kiwisolver>=1.3.1 in /Users/thomas200905/Documents/Thomas/HCMUS/Third Year/Semester 8/Data Visualisation/.venv/lib/python3.12/site-packages (from matplotlib) (1.4.9)

Requirement already satisfied: packaging>=20.0 in /Users/thomas200905/Documents/Thomas/HCMUS/Third Year/Semester 8/Data Visualisation/.venv/lib/python3.12/site-packages (from matplotlib) (25.0)

Requirement already satisfied: pillow>=8 in /Users/thomas200905/Documents/Thomas/HCMUS/Third Year/Semester 8/Data Visualisation/.venv/lib/python3.12/site-packages (from matplotlib) (12.1.0)

Requirement already satisfied: pyparsing>=3 in /Users/thomas200905/Documents/Thomas/HCMUS/Third Year/Semester 8/Data Visualisation/.venv/lib/python3.12/site-packages (from matplotlib) (3.3.1)

Requirement already satisfied: python-dateutil>=2.7 in /Users/thomas200905/Documents/Thomas/HCMUS/Third Year/Semester 8/Data Visualisation/.venv/lib/python3.12/site-packages (from matplotlib) (2.9.0.post0)

Requirement already satisfied: pytz>=2020.1 in /Users/thomas200905/Documents/Thomas/HCMUS/Third Year/Semester 8/Data Visualisation/.venv/lib/python3.12/site-packages (from pandas) (2025.2)

Requirement already satisfied: tzdata>=2022.7 in /Users/thomas200905/Documents/Thomas/HCMUS/Third Year/Semester 8/Data Visualisation/.venv/lib/python3.12/site-packages (from pandas) (2025.3)

Requirement already satisfied: scipy>=1.10.0 in /Users/thomas200905/Documents/Thomas/HCMUS/Third Year/Semester 8/Data Visualisation/.venv/lib/python3.12/site-packages (from scikit-learn) (1.17.1)

Requirement already satisfied: joblib>=1.3.0 in /Users/thomas200905/Documents/Thomas/HCMUS/Third Year/Semester 8/Data Visualisation/.venv/lib/python3.12/site-packages (from scikit-learn) (1.5.3)

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Requirement already satisfied: six>=1.5 in /Users/thomas200905/Documents/Thomas/HCMUS/Third Year/Semester 8/Data Visualisation/.venv/lib/python3.12/site-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)

LAB 2: DATA VISUALIZATION IN PYTHON

Course: Data Visualisation

Instructor: Vo Nhat Tan

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Introduction & Lab Objectives

Lab Purpose: This notebook processes and visualizes data from the **World Bank's World Development Indicators (WDI)**. Through 15 specific visualization objectives using Python, the goal is to uncover changing trends in macroeconomics, energy shifts, and human development across the Asia-Pacific region.

Tools & Libraries:

- **pandas** for time-series data wrangling and missing value handling.
- **matplotlib** & **seaborn** for rendering analytical charts.
- **scikit-learn** & **numpy** for scaling data (e.g., Radar charts) and complex multi-dimensional calculations.

1. Basic Data Analysis

1.1. Dataset Introduction

This dataset uses the **World Development Indicators (WDI)** collected from the World Bank.

Dataset Scope:

- **Timeframe:** 15 consecutive years from **2008 to 2022**.
- **Geography:** 16 countries located in the Asia-Pacific and ASEAN regions (e.g., Viet Nam, Japan, Australia, China, Singapore).
- **Indicators:** 13 variables divided carefully into 5 main groups. Below is the detailed breakdown of the fields we will be analyzing:

1. Macroeconomy:

- *Purpose:* To measure the overall health, stability, and growth of a country's economy.
- **GDP growth:** The annual percentage growth rate of GDP at market prices. (*Unit:* %)

- **Inflation, consumer prices:** The annual percentage change in the cost to the average consumer. (*Unit: %*)
- **Unemployment, total:** The share of the labor force that is without work but available for and seeking employment. (*Unit: % of total labor force*)

2. Energy Transition:

- *Purpose: To track how fast countries are moving away from fossil fuels and toward clean, sustainable power.*
- **Electricity production from oil, gas and coal sources:** Showcases the reliance on fossil fuels for electricity. (*Unit: % of total electricity*)
- **Electricity production from renewable sources, excluding hydroelectric:** Shows the shift towards green energy types like solar and wind. (*Unit: % of total electricity*)

3. Environment:

- *Purpose: To assess a country's direct environmental impact and contribution to global climate change.*
- **Carbon dioxide (CO2) emissions (excluding LULUCF):** Total CO2 gas emissions. (*Unit: Mt CO2e - Mega tonnes*)
- **Methane (CH4) emissions (excluding LULUCF):** Total CH4 gas emissions. (*Unit: % change from 1990*)

4. Human Development & Infrastructure:

- *Purpose: To evaluate the actual quality of life, basic living standards, and investment in future generations.*
- **Life expectancy at birth, total:** How many years a newborn is expected to live. (*Unit: years*)
- **School enrollment, tertiary:** The ratio of students enrolled in universities/colleges. (*Unit: % gross*)
- **Government expenditure on education, total:** How much a country's government spends on education. (*Unit: % of GDP*)
- **Access to electricity, rural:** The percentage of the rural population with electricity. (*Unit: % of rural population*)
- **People using at least basic drinking water services:** The percentage of people with safe water. (*Unit: % of population*)

5. Demographic:

- *Purpose: To provide scale. It allows us to calculate "per person" metrics to fairly compare countries with a large population like China with countries with a small population like Timor-Leste.*
- **Population, total:** Total count of all residents. (*Unit: People*)

Initial Inspection:

- The raw `data.csv` file has **213 rows** and **19 columns**.

- The first 4 columns contain IDs (Country Name, Country Code, Series Name, Series Code), and the next 15 columns cover the yearly data.
- The final 5 rows are empty metadata left over from the World Bank export.

First, we load the raw dataset to inspect its dimensions, data types, and identify any structural anomalies.

```
In [2]: import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

# 1. Load Data & Overview
df = pd.read_csv('WDI/data.csv')
print("Size:", df.shape)

df.info()

display(df.head())
display(df.tail())
```

```
Size: (213, 19)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 213 entries, 0 to 212
Data columns (total 19 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Country Name          210 non-null    object
1   Country Code          208 non-null    object
2   Series Name           208 non-null    object
3   Series Code           208 non-null    object
4   2008 [YR2008]         208 non-null    object
5   2009 [YR2009]         208 non-null    object
6   2010 [YR2010]         208 non-null    object
7   2011 [YR2011]         208 non-null    object
8   2012 [YR2012]         208 non-null    object
9   2013 [YR2013]         208 non-null    object
10  2014 [YR2014]         208 non-null    object
11  2015 [YR2015]         208 non-null    object
12  2016 [YR2016]         208 non-null    object
13  2017 [YR2017]         208 non-null    object
14  2018 [YR2018]         208 non-null    object
15  2019 [YR2019]         208 non-null    object
16  2020 [YR2020]         208 non-null    object
17  2021 [YR2021]         208 non-null    object
18  2022 [YR2022]         208 non-null    object
dtypes: object(19)
memory usage: 31.7+ KB
```

	Country Name	Country Code	Series Name	Series Code	2008 [YR2008]	2009 [YR2009]	2010 [YR2010]	2011 [YR2011]
0	Brunei Darussalam	BRN	Government expenditure on education, total (% ...)	SE.XPD.TOTL.GD.ZS	..	..	..	..
1	Brunei Darussalam	BRN	School enrollment, tertiary (% gross)	SE.TER.ENRR	15.7385101318359	16.565	..	..
2	Brunei Darussalam	BRN	Unemployment, total (% of total labor force) (...)	SL.UEM.TOTL.NE.ZS	..	..	..	..
3	Brunei Darussalam	BRN	Electricity production from oil, gas and coal ...	EG.ELC.FOSL.ZS	100	..	..	..
4	Brunei Darussalam	BRN	Electricity production from renewable sources,...	EG.ELC.RNWX.ZS	0	..	..	..
	Country Name	Country Code	Series Name	Series Code	2008 [YR2008]	2009 [YR2009]	2010 [YR2010]	2011 [YR2011]
208	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
209	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
210	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
211	Data from database: World Development Indicators	NaN	NaN	NaN	NaN	NaN	NaN	NaN
212	Last Updated: 02/24/2026	NaN	NaN	NaN	NaN	NaN	NaN	NaN

From the `df.info()` output and data preview, we found a few data quality issues:

1. **Footer rows:** The last 5 rows are export metadata with no actual data, which can interfere with filtering or calculations.
2. **Missing placeholders:** Missing values are marked as `..` instead of the standard `NaN`.
3. **Data types:** Because of the `..` text, pandas reads the year columns (`[YR2008]`, etc.) as text (`object`) instead of numbers (`float`).
4. **Wide format:** The years are spread out as separate columns. This "wide" structure makes it difficult to plot grouped charts later.

=> **Cleaning Plan:** To fix this, we will drop the last 5 rows, clean the column names, replace `..` with `NaN`, set the correct numeric data type, and use `melt` to restructure the data into a "long format".

1.2. Data Cleaning and Reshaping

The data is transformed in 3 steps:

1. **Filtering:** Remove the 5 footer rows and drop redundant identifier columns (`Series Code` , `Country Code`).
2. **Standardization:** Convert `2008 [YR2008]` style headers into plain years (`2008`), replace `..` with `NaN`, and cast all year values to numeric types.
3. **Reshaping (Melt):** Convert from wide format (one column per year) into a long format with fields: `Country Name` , `Series Name` , `Year` , `Value` . This format is essential for visualizations in Seaborn/Matplotlib.

We will remove footers, handle placeholders, standardize column names, and reshape the dataset for visualization.

```
In [3]: # Drop metadata footers
df_clean = df.dropna(subset=['Country Code']).copy()
df_clean = df_clean.drop(['Series Code', 'Country Code'], axis=1)

# Rename year columns ('2008 [YR2008]' -> '2008')
df_clean = df_clean.rename(columns={col: col.split(' ')[0] for col in df_

# Handle placeholders and cast to float
year_cols = sorted([col for col in df_clean.columns if col.isdigit()], ke
df_clean[year_cols] = df_clean[year_cols].replace('..', np.nan)
df_clean[year_cols] = df_clean[year_cols].astype(float)

# Melt to Long Format for visualization later
id_vars = ['Country Name', 'Series Name']
df_melted = pd.melt(df_clean, id_vars=id_vars, value_vars=year_cols, var_
df_melted['Year'] = df_melted['Year'].astype(int)

print(f"Rows (long format): {df_melted.shape[0]}")
print(f"Countries: {df_melted['Country Name'].nunique()} | Variables: {df_
print(f"Year range: {df_melted['Year'].min()}--{df_melted['Year'].max()} (

print("\n--- CLEANED DATA TYPES ---")
print(df_melted.dtypes)
```

```
Rows (long format): 3120
Countries: 16 | Variables: 13
Year range: 2008–2022 (15 years)
```

```
--- CLEANED DATA TYPES ---
Country Name    object
Series Name     object
Year            int64
Value           float64
dtype: object
```

- The dataset is completely transformed into **3,120 long records (16 countries x 13 indicators x 15 years)**.
- The structure now safely contains 4 standardized fields: `Country Name`, `Series Name`, `Year`, and `Value`.
- **Data Types:** `Year` is successfully cast to `int64` and the measurement `Value` is recognized as `float64`.

1.3. Missing Value Evaluation

Before determining an imputation strategy, we check the missing value counts across Series, Year, and Country.

```
In [4]: # Missing Value Evaluation
missing_counts = df_melted[df_melted['Value'].isnull()].groupby('Series Name').size()
total_counts = df_melted.groupby('Series Name').size()

missing_stats = pd.DataFrame({
    'Missing': missing_counts,
    'Expected': total_counts,
    'Missing (%)': (missing_counts / total_counts) * 100
}).fillna(0).sort_values(by='Missing (%)', ascending=False)
display(missing_stats)

print("\n--- MISSING BY YEAR ---")
print(df_melted[df_melted['Value'].isnull()].groupby('Year').size())

print("\n--- MISSING BY COUNTRY ---")
print(df_melted[df_melted['Value'].isnull()].groupby('Country Name').size())
```


	Missing	Expected	Missing (%)
Series Name			
Government expenditure on education, total (% of GDP)	71.0	240	29.583333
School enrollment, tertiary (% gross)	53.0	240	22.083333
Unemployment, total (% of total labor force) (national estimate)	44.0	240	18.333333
Electricity production from renewable sources, excluding hydroelectric (% of total)	16.0	240	6.666667
Inflation, consumer prices (annual %)	3.0	240	1.250000
Electricity production from oil, gas and coal sources (% of total)	1.0	240	0.416667
Access to electricity, rural (% of rural population)	0.0	240	0.000000
Carbon dioxide (CO2) emissions (total) excluding LULUCF (Mt CO2e)	0.0	240	0.000000
GDP growth (annual %)	0.0	240	0.000000
Life expectancy at birth, total (years)	0.0	240	0.000000
Methane (CH4) emissions (total) excluding LULUCF (% change from 1990)	0.0	240	0.000000
People using at least basic drinking water services (% of population)	0.0	240	0.000000
Population, total	0.0	240	0.000000

--- MISSING BY YEAR ---

Year

2008	18
2009	16
2010	12
2011	13
2012	15
2013	14
2014	11
2015	10
2016	11
2017	7
2018	9
2019	7
2020	9
2021	8
2022	28

dtype: int64

--- MISSING BY COUNTRY ---

Country Name

Myanmar	36
Timor-Leste	25
Australia	20
Brunei Darussalam	19
China	19

dtype: int64

Missing Value Evaluation: In total, there are **188 missing records**.

- **By Series:** Missing data is highest in *Government expenditure on education* (71), *Tertiary enrollment* (53), and *Unemployment* (44). Public spending and labor data seem harder to track consistently.
- **By Year:** **2022** has the most missing data (28 records). Recent data updates often take time to be published by the World Bank.
- **By Country:**
 - Developing nations like **Myanmar** (36) and **Timor-Leste** (25) lack data mainly due to limited statistical systems or national instability.
 - Developed nations like **Australia** (20) also miss data, but mostly in education metrics. This means their internal reporting standards don't match the World Bank's definitions.

1.4. Handling Missing Data

Method: Linear Interpolation (`method='linear'`). Filling missing values with an overall mean or carrying the last value forward would create flat, unnatural lines in our charts. Linear interpolation connects the dots between two known years to keep the trend smooth.

We used `limit=4` and `limit_area='inside'` .

- It only fills small internal gaps (up to 4 consecutive years).
- It **does not** guess data at the edges. If a country has no data for 2022 at the end of the line, it correctly stays as **NaN** .

```
In [5]: # Sort the melted DataFrame for interpolation
df_melted = df_melted.sort_values(by=['Country Name', 'Series Name', 'Year'])
df_processed = df_melted.reset_index(drop=True).copy()

# Interpolate max 4 consecutive years, inside existing data points (some
df_processed['Value'] = df_processed.groupby(['Country Name', 'Series Name'])
    .lambda x: x.interpolate(method='linear', limit=4, limit_area='inside')
    .reset_index()

print("Total missing values before interpolation:", df_melted['Value'].isna().sum())
print("Total missing values after interpolation:", df_processed['Value'].isna().sum())
```

Total missing values before interpolation: 188

Total missing values after interpolation: 138

Result: Missing records were safely reduced from **188 to 138**. We filled exactly **50 internal gaps** between years but kept the endpoint gaps honest.

1.5. Descriptive Statistics

With the dataset cleaned and interpolated, we pivot to compute descriptive statistics (count, mean, min, max, standard deviation) across all **13 indicators**.

```
In [6]: pivot_df = df_processed.pivot_table(index=['Country Name', 'Year'],
                                             columns='Series Name', values='Value')

print("STATISTICS")
display(pivot_df.describe().T[['count', 'mean', 'min', 'max', 'std']])
```

STATISTICS

	count	mean	min	max	std
Series Name					
Year	240.0	2.015000e+03	2008.000000	2.022000e+03	4.329523e+00
Access to electricity, rural (% of rural population)	240.0	8.840792e+01	11.700000	1.000000e+02	2.104462e+01
Carbon dioxide (CO2) emissions (total) excluding LULUCF (Mt CO2e)	240.0	9.004619e+02	0.310900	1.255186e+04	2.568095e+03
Electricity production from oil, gas and coal sources (% of total)	239.0	7.058465e+01	0.000000	1.000000e+02	2.644896e+01
Electricity production from renewable sources, excluding hydroelectric (% of total)	224.0	6.247243e+00	-9.996215	2.711663e+01	7.131819e+00
GDP growth (annual %)	240.0	4.084775e+00	-20.522479	3.172574e+01	4.597826e+00
Government expenditure on education, total (% of GDP)	183.0	3.583144e+00	0.787440	1.290248e+01	1.823713e+00
Inflation, consumer prices (annual %)	237.0	3.227385e+00	-1.610680	2.679954e+01	3.903966e+00
Life expectancy at birth, total (years)	240.0	7.450206e+01	56.364000	8.456000e+01	6.552374e+00
Methane (CH4) emissions (total) excluding LULUCF (% change from 1990)	240.0	7.086318e+01	-24.707796	3.946733e+02	8.158562e+01
People using at least basic drinking water services (% of population)	240.0	9.077415e+01	59.681726	1.000000e+02	1.119716e+01

	count	mean	min	max	std
Series Name					
Population, total	240.0	1.388928e+08	377793.000000	1.412360e+09	3.267587e+08
School enrollment, tertiary (% gross)	200.0	4.639803e+01	8.929950	1.196828e+02	3.010398e+01
Unemployment, total (% of total labor force) (national estimate)	219.0	3.354342e+00	0.141000	9.316000e+00	1.796704e+00

Descriptive Statistics Findings: With the data cleaned and gap-filled, we computed count, mean, min, max, and std:

1. **Different Scales:** The indicators have widely different ranges. For example, *GDP growth* is usually single digits (mean ~4.1%), but *Population* scales up to 1.41 billion. So when plotting, we need to use separate charts, dual-axes, or normalize the data.
2. **Economic Volatility:** Economic markers are unstable. *GDP growth* ranges from a -20.5% drop to a 31.7% surge. *Inflation* also fluctuates from -1.6% to 26.8%.
3. **Huge Infrastructure Gaps:** Developed countries usually have 100% access, but developing countries drag the minimums down. *Rural electricity access* drops to 11.7%, and *Basic drinking water access* is 59.7% at its worst.
4. **Total vs. Per Capita Bias:** Absolute amounts like *Total CO2 Emissions* vary hugely (from 0.31 Mt to 12,551.8 Mt) just because of country size. We will need to look at per-capita metrics to compare efficiency fairly.

Summary

Our pre-processing is complete, and the data is ready for visualization:

1. **Clean Structure:** Tidy long-format dataset of **3,120 rows** (`Country Name` , `Series Name` , `Year` , `Value`), covering 16 countries and 13 indicators over 15 years.
2. **Missing Data:** Interpolation filled 50 short gaps, leaving exactly **138 missing records** at the edges.
3. **Plotting Strategy:** Variables like Population and percentages exist on different scales, we must be careful to use separate charts or dual-axes.

2. Objectives and Feature Selection (Task 2)

This section outlines 15 analytical objectives across 5 socio-economic themes. For each objective, we specify the target metrics, their rationale, and the chosen visualization technique focusing on data properties.

Theme 1: Macroeconomic Trends and Stability

Objective 1: Track GDP growth over time (2008-2022)

- **Metric:** GDP growth (annual %).
- **Meaning:** Year-over-year economic growth rate.
- **Rationale:** Standard indicator to measure continuous economic expansion or contraction.
- **Chart choice:** Line chart. Ideal for showing continuous trends over time for time-series data.

Objective 2: Examine the inflation-unemployment relationship (Phillips Curve)

- **Metrics:** Inflation (annual %) & Unemployment (% of labor force).
- **Meaning:** Measures price increases against job shortages.
- **Rationale:** Tests the Phillips Curve theory (the inverse relationship between inflation and unemployment).
- **Chart choice:** Scatter plot with trendline. Perfect for showing the correlation between two numerical variables.

Objective 3: Compare GDP growth stability and distribution shape

- **Metric:** GDP growth (annual %).
 - **Meaning:** Distribution of yearly growth outcomes by country.
 - **Rationale:** Stability is a distribution question, not only an average question.
 - **Chart choice:** Boxplot, suitable for comparing median, spread, and outliers across categories.
-

Theme 2: Energy Transition and Environmental Impact

Objective 4: Assess the shift from fossil to renewable electricity

- **Metrics:** Electricity from fossil fuels (%) & from renewables (%).
- **Meaning:** Percentage breakdown of electricity sources.
- **Rationale:** Directly tracks a country's transition toward clean energy.
- **Chart choice:** Stacked area chart. Best for showing both the part-to-whole relationship (composition) and how it evolves over continuous time.

Objective 5: Evaluate progress in Methane (CH₄) emission mitigation

- **Metric:** Methane emissions (% change from 1990).
- **Meaning:** Relative change in methane emissions compared to the 1990 baseline.
- **Rationale:** Percentage change enables fair comparisons regardless of country size.

- **Chart choice:** Horizontal bar chart. Most effective for ranking categorical data measured against a single numerical value.

Objective 6: Track CO2 emissions trend and efficiency (2008-2022)

- **Metrics:** Total CO2 emissions (Mt CO2e) & CO2 per capita (Tonnes).
 - **Meaning:** Overall emission scale vs. individual carbon footprint.
 - **Rationale:** Contrasts total national impact with per-person environmental efficiency.
 - **Chart choice:** Two side-by-side line charts. Useful for side-by-side comparisons of time-series data at different scales (absolute vs. normalized per capita ratios).
-

Theme 3: Human Capital and Educational Investment

Objective 7: Evaluate government priority for education

- **Metric:** Government expenditure on education (% of GDP).
- **Meaning:** Share of national wealth invested in public education.
- **Rationale:** Standardizes spending by economic size to fairly compare policy priorities.
- **Chart choice:** Horizontal bar chart. Most effective for ranking categorical data (countries) based on a single numerical value.

Objective 8: Examine education investment vs. tertiary enrollment

- **Metrics:** Education expenditure (% of GDP) & Tertiary enrollment (% gross).
- **Meaning:** Financial input versus educational outcome.
- **Rationale:** Tests whether higher financial investment correlates with higher university enrollment.
- **Chart choice:** Scatter plot. Effectively maps the linear relationship between financial input (x-axis) and educational output (y-axis) across numerical data points.

Objective 9: Monitor expansion of tertiary education access

- **Metric:** Tertiary enrollment (% gross).
 - **Meaning:** Percentage of the population attending higher education.
 - **Rationale:** Tracks the long-term development of a highly educated workforce.
 - **Chart choice:** Line chart. Ideal for comparing multi-year growth trajectories for time-series data across categorical entities (nations).
-

Theme 4: Infrastructure and Public Health

Objective 10: Measure progress in rural electrification

- **Metric:** Access to electricity, rural (% of rural population).
- **Meaning:** Coverage of electricity in rural demographics.
- **Rationale:** A core indicator of infrastructure development and poverty reduction.

- **Chart choice:** Line chart. Highlights infrastructural catch-up progress for continuous time-series data, specifically focusing on developing nations to contrast with a developed nation that has maintained full coverage.

Objective 11: Evaluate water access and life expectancy relationship

- **Metrics:** Basic drinking water access (%) & Life expectancy at birth (years).
- **Meaning:** Basic public sanitation coverage versus human lifespan.
- **Rationale:** Verifies the developmental link between basic infrastructure and health outcomes.
- **Chart choice:** Scatter plot. Clearly reveals clusters and positive correlations between two continuous numerical variables in global public health data.

Objective 12: Compare the distribution of life expectancy across countries

- **Metric:** Life expectancy at birth (years).
- **Meaning:** The expected average lifespan.
- **Rationale:** Summarizes overall health quality and highlights the global health gap.
- **Chart choice:** Violin plot. Ideal for comparing the probability distribution and central tendency (median/quartiles) of numerical data across multiple categories.

Theme 5: Multi-Dimensional Synthesis

Objective 13: Build a cross-indicator correlation overview

- **Metrics:** All 13 numeric indicators.
- **Meaning:** Pairwise relationships among all variables.
- **Rationale:** Provides a system-level scan for macro-patterns before specific analysis.
- **Chart choice:** Correlation heatmap. Provides a global view of pairwise correlation coefficients across all continuous numerical variables in a single matrix.

Objective 14: Compare multi-dimensional country profiles

- **Metrics:** GDP growth, CO2 per capita, Tertiary enrollment, Access to electricity, Life expectancy.
- **Meaning:** A holistic snapshot across economic, environmental, educational, and health pillars.
- **Rationale:** Evaluates a country's overall development balance.
- **Chart choice:** Radar chart (Spider chart). Perfect for synthesizing multi-dimensional numerical data into a profile to compare categorical entities (countries). *(Requires Min-Max scaling to align different units).*

Objective 15: Explore the modernization trade-off (Quality of Life vs. Emissions)

- **Metrics:** Life expectancy (Y-axis) and CO2 per capita (X-axis).
- **Meaning:** Health outcomes versus individual carbon cost.
- **Rationale:** Examines if living longer necessitates higher emissions, comparing individual environmental efficiencies across different levels of modernization.
- **Chart choice:** Scatter plot. Directly maps and highlights the relationship between two continuous numerical variables to identify efficient versus inefficient models.

3. Visualization and Analysis

Theme 1: Macroeconomic Trends and Stability

Objective 1: Track GDP growth over time (2008-2022)

- **Chart Choice:** Line chart. Ideal for showing continuous trends over time for time-series data.
- **Design Choice:** Plotting 16 countries simultaneously can create a confusing "spaghetti plot". We will highlight a few key economies (e.g., Vietnam, China, Australia, Japan) and fade the rest into the background to make the trends readable.

```
In [7]: plt.figure(figsize=(12, 6))

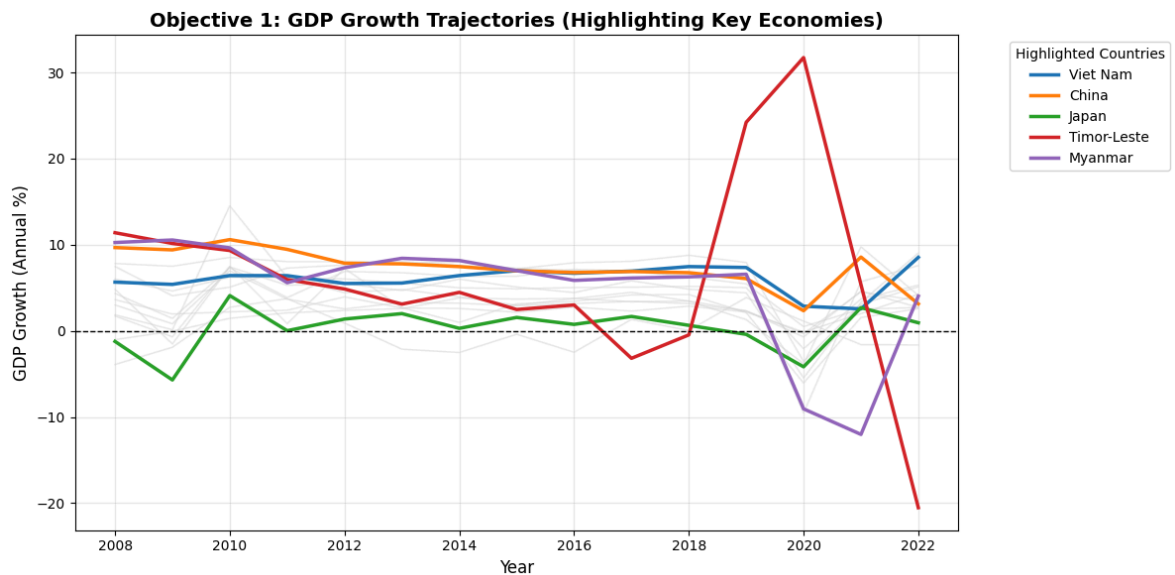
# Define countries to highlight
# highlight_countries = ['Viet Nam', 'China', 'Australia', 'Japan', 'Kore
#                        'Thailand', 'Singapore', 'Philippines', 'Myanmar'
highlight_countries = ['Viet Nam', 'China', 'Japan', 'Timor-Leste', 'Myan
palette = sns.color_palette("tab10", len(highlight_countries))

for country in pivot_df['Country Name'].unique():
    if country not in highlight_countries:
        country_data = pivot_df[pivot_df['Country Name'] == country]
        sns.lineplot(data=country_data, x='Year', y='GDP growth (annual %'

for idx, country in enumerate(highlight_countries):
    country_data = pivot_df[pivot_df['Country Name'] == country]
    sns.lineplot(data=country_data, x='Year', y='GDP growth (annual %)',

plt.axhline(0, color='black', linestyle='--', linewidth=1)

plt.title('Objective 1: GDP Growth Trajectories (Highlighting Key Economi
plt.xlabel('Year', fontsize=12)
plt.ylabel('GDP Growth (Annual %)', fontsize=12)
plt.legend(title='Highlighted Countries', bbox_to_anchor=(1.05, 1), loc='
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```



Analysis & Conclusion (Objective 1):

- **General Trend:** Most economies in the dataset grow at 10% or less per year, with occasional years of negative growth (economic contraction).
- **High Instability (Timor-Leste):** Timor-Leste is the most volatile country on the chart. Its growth dropped from 11% in 2008 to 0% in 2018, spiked to 30% in 2020, and then crashed to -20% in 2022.
- **Stable Growth (Viet Nam):** Viet Nam shows highly stable economic performance, consistently maintaining a growth rate of around 5% to 7% per year before the pandemic.
- **Stagnation (Japan):** Japan displays economic stagnation. Its annual growth rate hovers near 0% and drops into negative numbers multiple times over the 15-year period.
- **Slowing Growth (China):** China's economy is relatively stable but shows a clear, gradual slowdown over time, dropping from about 10% growth in 2008 down to roughly 3-5% by 2022.
- **Sudden Decline (Myanmar):** Myanmar maintained positive growth until 2019. The economy then shrank severely over the next two years (dropping to nearly -12%) before rebounding slightly in 2022.

Objective 2: Examine the inflation-unemployment relationship (Phillips Curve)

- **Chart Choice:** Scatter plot with a regression line. Perfect for showing the correlation between two continuous numerical variables to test economic theories.

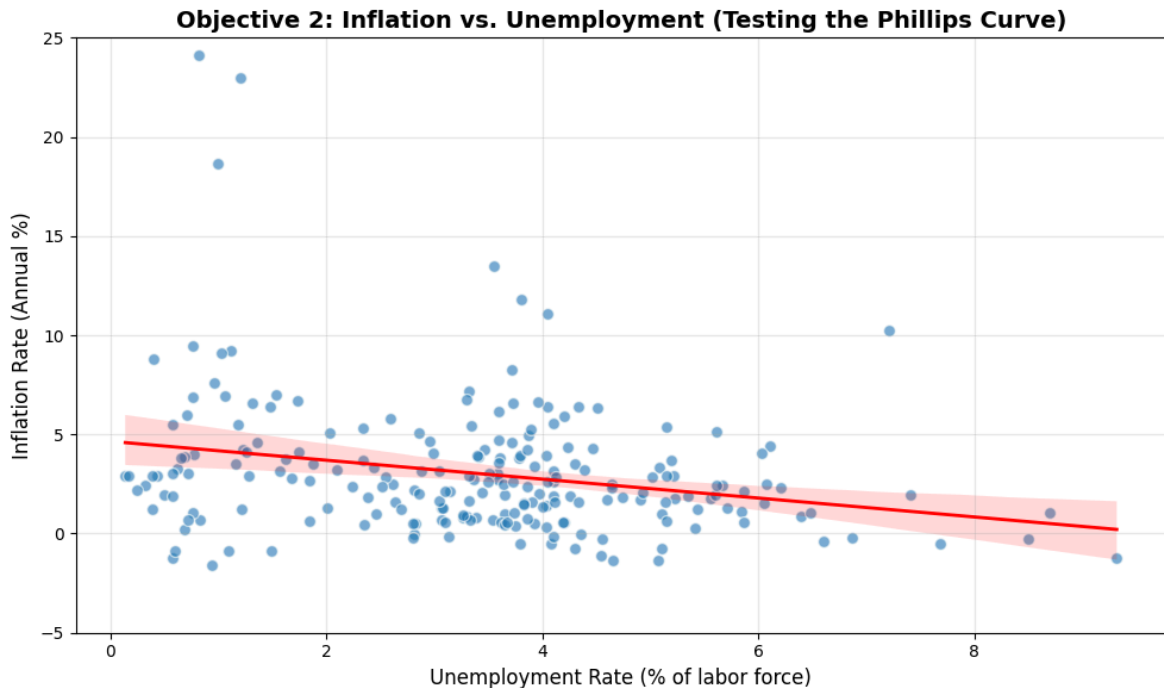
```
In [8]: plt.figure(figsize=(10, 6))

sns.regplot(data=pivot_df,
            x='Unemployment, total (% of total labor force) (national est
            y='Inflation, consumer prices (annual %)',
            scatter_kws={'alpha': 0.6, 'edgecolor': 'w', 's': 50},
            line_kws={'color': 'red', 'linewidth': 2})

plt.title('Objective 2: Inflation vs. Unemployment (Testing the Phillips
```

```
plt.xlabel('Unemployment Rate (% of labor force)', fontsize=12)
plt.ylabel('Inflation Rate (Annual %)', fontsize=12)

plt.ylim(-5, 25)
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```



Analysis & Conclusion (Objective 2):

- **Inverse Relationship:** The graph shows a general inverse relationship, meaning higher inflation correlates with lower unemployment. When unemployment is very low (around 1%), inflation averages near 5%. When unemployment is high (around 8%), inflation drops close to 0%.
- **Weak Correlation:** Even with a visible trend, this relationship is very weak. The data points are scattered far away from the red trendline. For example, at a 4% unemployment rate, inflation varies drastically across different countries, ranging anywhere from -2% to over 14%.

Objective 3: Compare GDP growth stability across countries

- **Chart Choice:** Boxplot. A boxplot excellently summarizes the distribution, central tendency (median), and volatility (IQR and outliers) of GDP growth for each country over the 15-year period.

```
In [9]: plt.figure(figsize=(12, 8))

highlight_countries = ['Viet Nam', 'China', 'Japan', 'Timor-Leste', 'Myan
palette = sns.color_palette("tab10", len(highlight_countries))

color_map = {country: '#e0e0e0' for country in pivot_df['Country Name'].u
for i, country in enumerate(highlight_countries):
    color_map[country] = palette[i]

median_gdp = pivot_df.groupby('Country Name')['GDP growth (annual %)].me
```

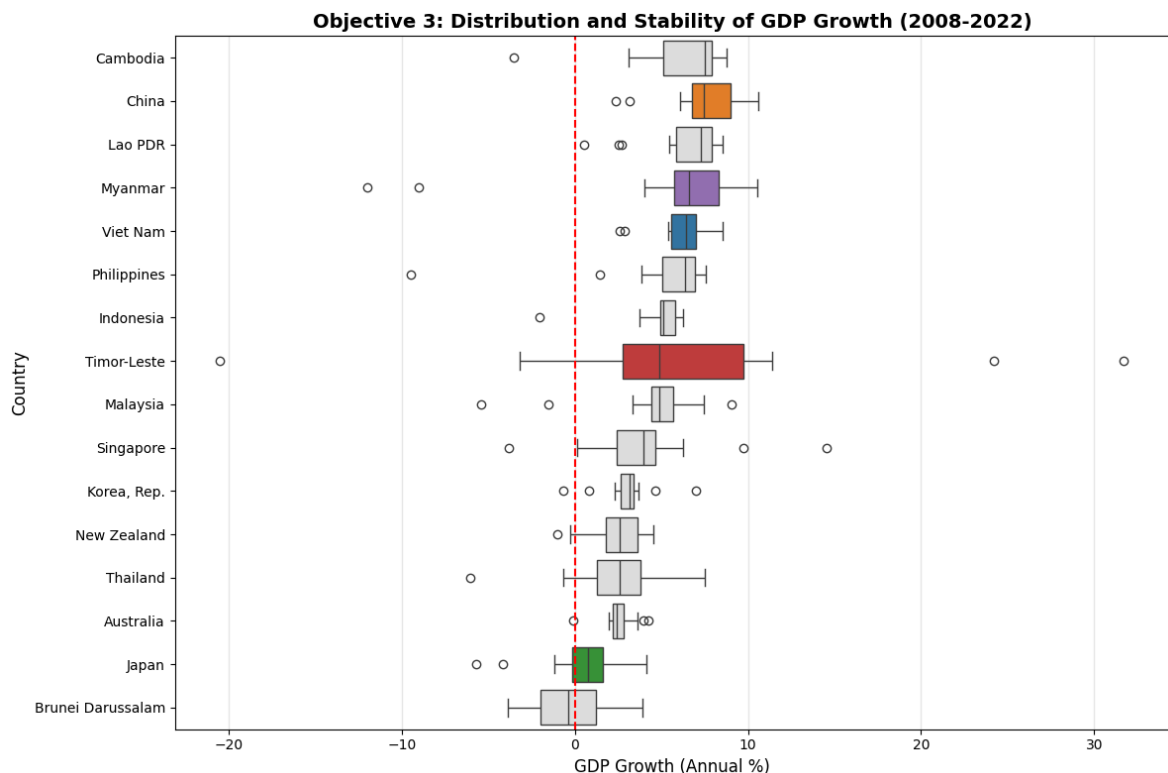
```

sns.boxplot(data=pivot_df,
            x='GDP growth (annual %)',
            y='Country Name',
            order=median_gdp.index,
            hue='Country Name',
            palette=color_map,
            legend=False)

plt.axvline(0, color='red', linestyle='--', linewidth=1.5)

plt.title('Objective 3: Distribution and Stability of GDP Growth (2008-2022)')
plt.xlabel('GDP Growth (Annual %)', fontsize=12)
plt.ylabel('Country', fontsize=12)
plt.grid(axis='x', alpha=0.3)
plt.tight_layout()
plt.show()

```



Analysis & Conclusion (Objective 3):

- **Overall Growth vs. Stagnation:** Most countries have a positive median growth rate, meaning their economies generally grew over the period. Japan shows a median close to 0%, confirming its long-term economic stagnation.
- **Stable Performers (Viet Nam & China):** Viet Nam and China rank highly for median growth (around 5-7%) and have very narrow boxes. This confirms they had consistent, rapid economic expansion with minimal fluctuation over the entire period.
- **High Instability (Timor-Leste):** Timor-Leste's box and whiskers are extremely wide, and it has outliers reaching above 30% and below -20%, confirming it was the most unstable economy over the 15 years.
- **Short-term Decline (Myanmar):** Myanmar has a narrow box, indicating its economy was mostly stable. However, it has severe outlier points around -10%

and -12%, confirming a significant economic decline for a short time.

Theme 2: Energy Transition and Environmental Impact

Objective 4: Assess the shift from fossil to renewable electricity

- **Chart Choice:** Stacked Area Chart (Regional Average). Best for showing both the part-to-whole relationship (composition) and how it evolves over continuous time.

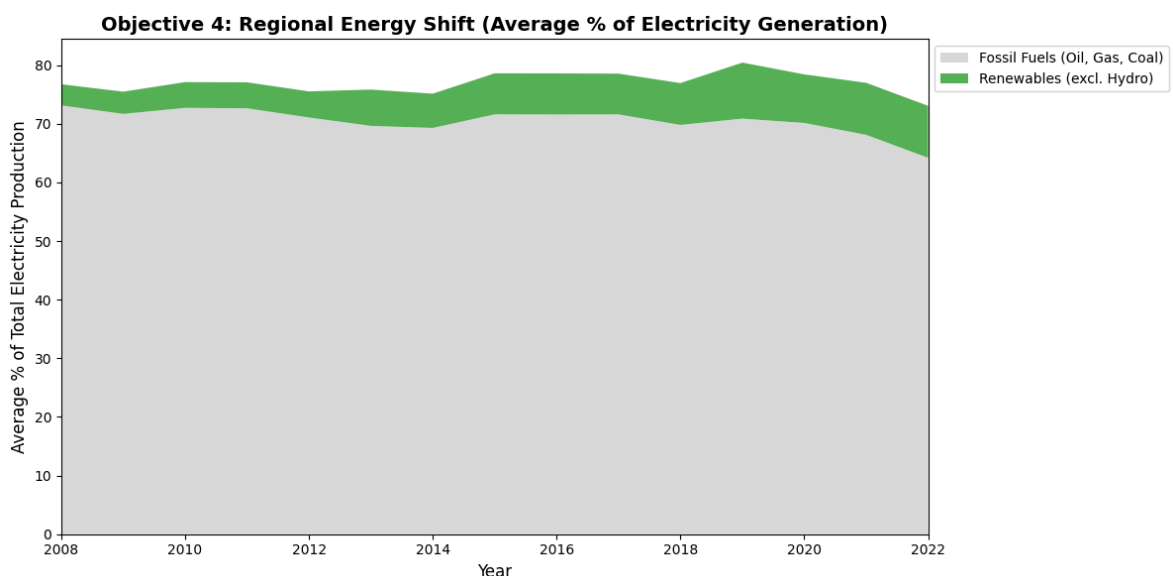
```
In [ ]: plt.figure(figsize=(12, 6))

# Calculate regional average per year
energy_trends = pivot_df.groupby('Year')[['Electricity production from oil, gas and coal',
                                           'Electricity production from renewable sources']]

# Fill missing values by forward filling (assuming trends continue in the future)
energy_trends.ffmpeg(inplace=True)

plt.stackplot(energy_trends.index,
              energy_trends['Electricity production from oil, gas and coal'],
              energy_trends['Electricity production from renewable sources'],
              labels=['Fossil Fuels (Oil, Gas, Coal)', 'Renewables (excl. Hydro)'],
              colors=['#d3d3d3', '#2ca02c'], alpha=0.8)

plt.title('Objective 4: Regional Energy Shift (Average % of Electricity Generation)')
plt.xlabel('Year', fontsize=12)
plt.ylabel('Average % of Total Electricity Production', fontsize=12)
plt.legend(loc='upper left', bbox_to_anchor=(1, 1))
plt.margins(x=0)
plt.tight_layout()
plt.show()
```



Analysis & Conclusion (Objective 4):

- **Fossil Fuel Dominance:** Fossil fuels remain the primary electricity source for the region. However, their regional average share decreased significantly from approximately 73% in 2008 to 64% by 2022.

- **Renewable Energy Growth:** The transition to renewable sources (excluding hydroelectric) shows steady and promising progress. Over the 15-year period, its regional average share more, growing from roughly 4% in 2008 to 8% by 2022.

Objective 5: Evaluate progress in Methane (CH₄) emission mitigation

- **Chart Choice:** Horizontal bar chart. Most effective for ranking categorical data measured against a single numerical value. Using red for positive numbers (emission increase) and green for negative numbers (emission decrease) makes the progress instantly clear.

```
In [ ]: plt.figure(figsize=(10, 8))

# Filter data for the most recent year (2022) to compare current progress
col_name = 'Methane (CH4) emissions (total) excluding LULUCF (% change fr
df_methane = pivot_df[pivot_df['Year'] == 2022].dropna(subset=[col_name])

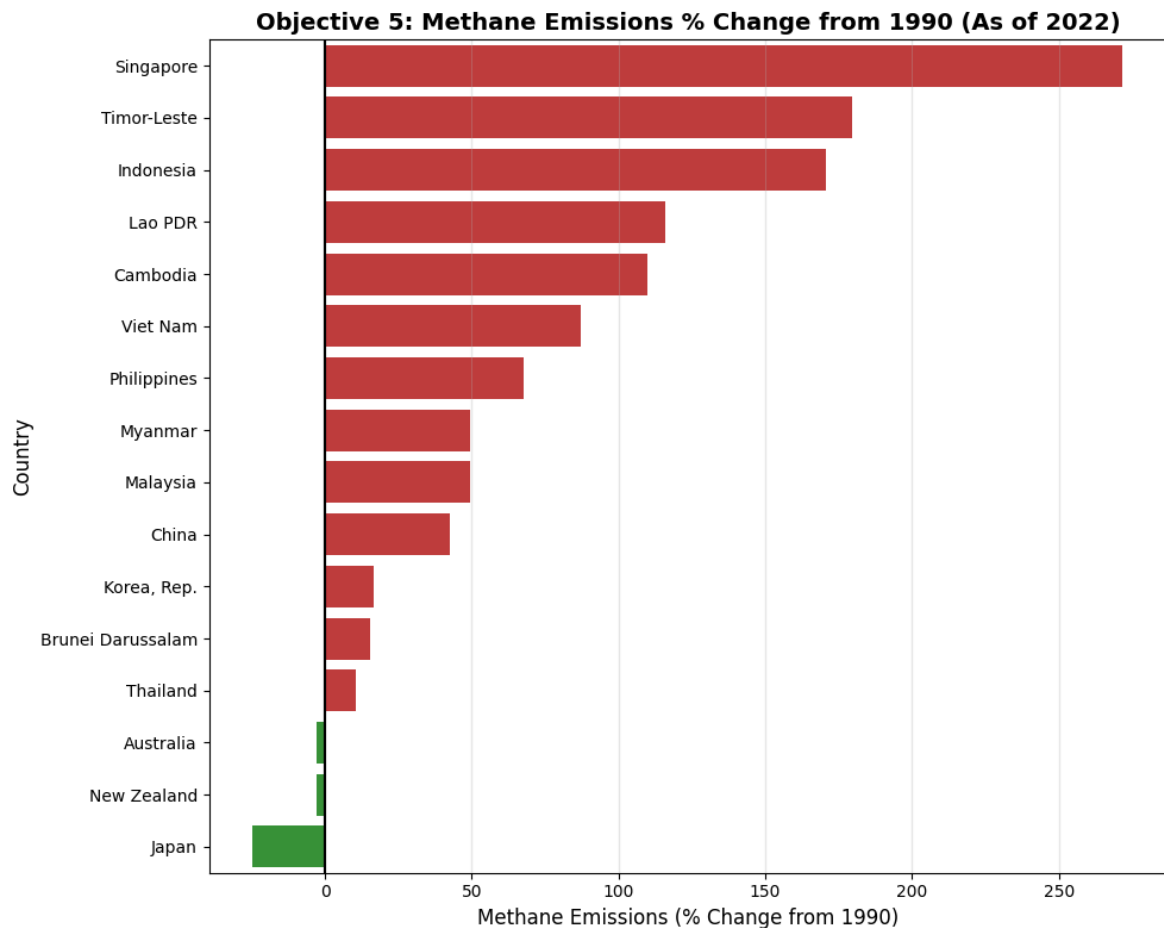
# Sort values for a clean bar chart
df_methane = df_methane.sort_values(by=col_name, ascending=False)

# Create a color palette: Red for increase (>0), Green for decrease (<0)
colors = ['#d62728' if x > 0 else '#2ca02c' for x in df_methane[col_name]]

# Updated barplot call to remove the deprecation warning
sns.barplot(data=df_methane,
            x=col_name,
            y='Country Name',
            hue='Country Name', # Assigning y variable to hue fixes the w
            palette=colors,      # Uses your custom red/green logic
            legend=False)       # Hides the redundant legend

# Add a vertical reference line at 0% (the 1990 baseline)
plt.axvline(0, color='black', linestyle='-', linewidth=1.5)

plt.title('Objective 5: Methane Emissions % Change from 1990 (As of 2022)')
plt.xlabel('Methane Emissions (% Change from 1990)', fontsize=12)
plt.ylabel('Country', fontsize=12)
plt.grid(axis='x', alpha=0.3)
plt.tight_layout()
plt.show()
```



Analysis & Conclusion (Objective 5):

- **Large Increases:** Singapore, Timor-Leste, Indonesia, Lao PDR, and Cambodia show large increases in methane emissions, all growing by over 100% since 1990. Singapore shows the highest increase at nearly 270%.
- **Moderate Growth:** Viet Nam, the Philippines, Myanmar, Malaysia, and China show moderate growth, increasing their methane emissions by roughly 40% to 85% compared to 1990 levels.
- **Reductions:** Only three countries—Japan, Australia, and New Zealand reduced their methane emissions below their 1990 levels.

Objective 6: Track CO2 emissions trend and efficiency (2008-2022)

- **Chart Choice:** Two side-by-side line charts. Useful for side-by-side comparisons of time-series data at different scales (absolute vs. normalized per capita ratios). The first chart uses a log-scaled Y-axis because the total emissions gap between massive countries (like China) and small ones (like Timor-Leste) is too big for a normal scale. The second chart shows emissions per person on a normal scale, which allows for a fair, direct comparison.

```
In [ ]: # Calculate CO2 emissions per capita (in Tonnes)
# Note: 1 Mt (Mega tonne) = 1,000,000 Tonnes
pivot_df['CO2 per capita (Tonnes)'] = (pivot_df['Carbon dioxide (CO2) emi

# Create a figure with 2 subplots side-by-side
fig, axes = plt.subplots(1, 2, figsize=(16, 6))
```

```

# Define key countries to highlight based on our analysis
highlight_countries = ['China', 'Timor-Leste', 'Lao PDR', 'Japan', 'Viet
palette = sns.color_palette("tab10", len(highlight_countries))

# --- Chart 1: Total CO2 Emissions (Log Scale) ---
# Plot background countries (light gray)
for country in pivot_df['Country Name'].unique():
    if country not in highlight_countries:
        country_data = pivot_df[pivot_df['Country Name'] == country]
        sns.lineplot(data=country_data, x='Year', y='Carbon dioxide (CO2)
                        color='lightgray', alpha=0.5, linewidth=1, ax=axes[0])

# Plot highlighted countries
for idx, country in enumerate(highlight_countries):
    country_data = pivot_df[pivot_df['Country Name'] == country]
    sns.lineplot(data=country_data, x='Year', y='Carbon dioxide (CO2) emi
                  color=palette[idx], linewidth=2.5, ax=axes[0])

axes[0].set_yscale('log')
axes[0].set_title('Total CO2 Emissions by Country (Log Scale)', fontsize=
axes[0].set_xlabel('Year', fontsize=12)
axes[0].set_ylabel('Total CO2 Emissions (Mt CO2e)', fontsize=12)
axes[0].grid(True, alpha=0.3)

# --- Chart 2: CO2 Emissions Per Capita ---
# Plot background countries (light gray)
for country in pivot_df['Country Name'].unique():
    if country not in highlight_countries:
        country_data = pivot_df[pivot_df['Country Name'] == country]
        sns.lineplot(data=country_data, x='Year', y='CO2 per capita (Tonnes
                        color='lightgray', alpha=0.5, linewidth=1, ax=axes[1])

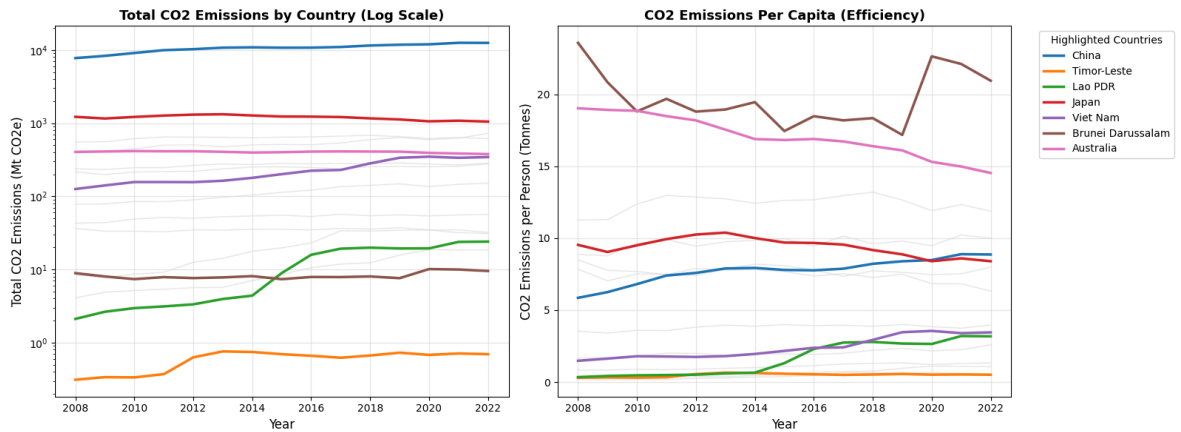
# Plot highlighted countries
for idx, country in enumerate(highlight_countries):
    country_data = pivot_df[pivot_df['Country Name'] == country]
    sns.lineplot(data=country_data, x='Year', y='CO2 per capita (Tonnes)'
                  color=palette[idx], label=country, linewidth=2.5, ax=axe

axes[1].set_title('CO2 Emissions Per Capita (Efficiency)', fontsize=13, f
axes[1].set_xlabel('Year', fontsize=12)
axes[1].set_ylabel('CO2 Emissions per Person (Tonnes)', fontsize=12)
axes[1].grid(True, alpha=0.3)

# Legend
axes[1].legend(title='Highlighted Countries', bbox_to_anchor=(1.05, 1), l

plt.tight_layout()
plt.show()

```

Analysis & Conclusion (Objective 6)

- Total Emissions (Log Scale):** Total emissions generally increased across the region. China is the dominant emitter (exceeding 10,000 Mt), while Timor-Leste remains the lowest (under 1 Mt). Lao PDR displays the steepest growth trajectory, particularly after 2014. Japan's emissions remained stable at roughly 1,200 Mt. Viet Nam experienced continuous growth, rising from approximately 100 Mt to near 400 Mt.
- Per Capita Efficiency:** Viewing the data relative to population reveals a different dynamic. Brunei Darussalam has the highest per capita emissions, fluctuating between 18 and 23 tonnes per person. Australia ranks second but shows a notable decline from 19 to 15 tonnes per person. China's per capita emissions (rising from 6 to 9 tonnes) remain lower than several smaller nations. Timor-Leste maintains the smallest per capita footprint (under 1 tonne per person).

Theme 3: Human Capital and Educational Investment

Objective 7: Evaluate government priority for education

- Chart Choice:** Horizontal Bar Chart. Most effective for ranking categorical data (countries) based on a single numerical value (average expenditure). Sorting the bars makes the comparison instantly readable.

```
In [ ]: plt.figure(figsize=(10, 8))

# Calculate average education expenditure per country over the available
edu_expenditure = pivot_df.groupby('Country Name')['Government expenditure']

# Drop countries with entirely NaN values for this metric to clean up the
edu_expenditure = edu_expenditure.dropna()

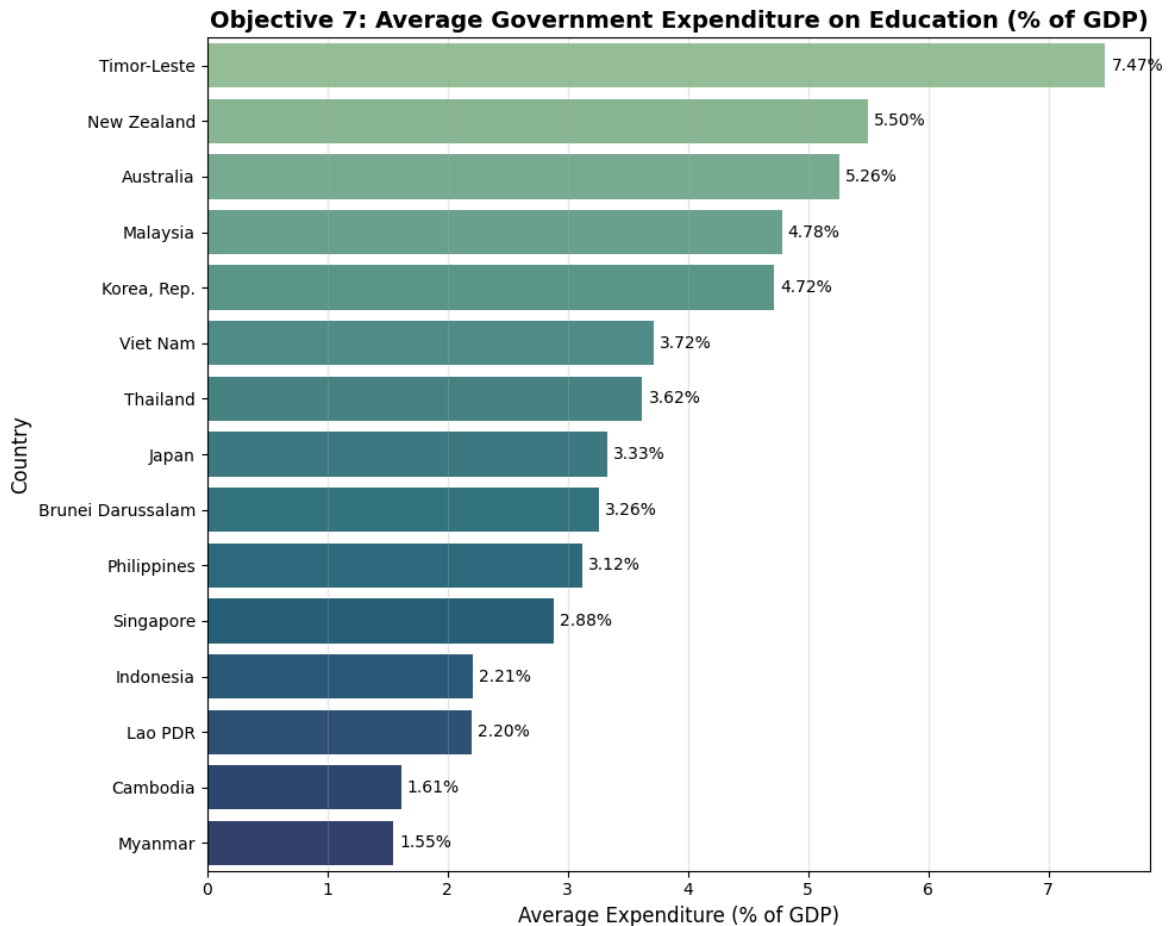
# Updated barplot call to remove the deprecation warning
sns.barplot(x=edu_expenditure.values,
            y=edu_expenditure.index,
            hue=edu_expenditure.index, # Assigning y variable to hue
            palette='crest',
            legend=False)                # Hiding the legend

plt.title('Objective 7: Average Government Expenditure on Education (% of
```

```
plt.xlabel('Average Expenditure (% of GDP)', fontsize=12)
plt.ylabel('Country', fontsize=12)

# Add value labels to the end of the bars
for index, value in enumerate(edu_expenditure.values):
    plt.text(value + 0.05, index, f'{value:.2f}%', va='center', fontsize=12)

plt.grid(axis='x', alpha=0.3)
plt.tight_layout()
plt.show()
```



Analysis & Conclusion (Objective 7):

- **Top Spender (Timor-Leste):** Timor-Leste spends the most on education, averaging 7.47% of its GDP over the period.
- **High Spenders (New Zealand & Australia):** New Zealand and Australia follow, both spending over 5% of their GDP on education (5.50% and 5.26%).
- **Middle Group:** The majority of the countries, including Viet Nam, Thailand, and Japan, spend between 3% and 4.7% of their GDP on education.
- **Lowest Spenders (Myanmar & Cambodia):** Myanmar (1.55%) and Cambodia (1.61%) spend the least on education. The gap between the highest and lowest spending countries is very wide (nearly 6%).

Objective 8: Examine education investment and tertiary enrollment

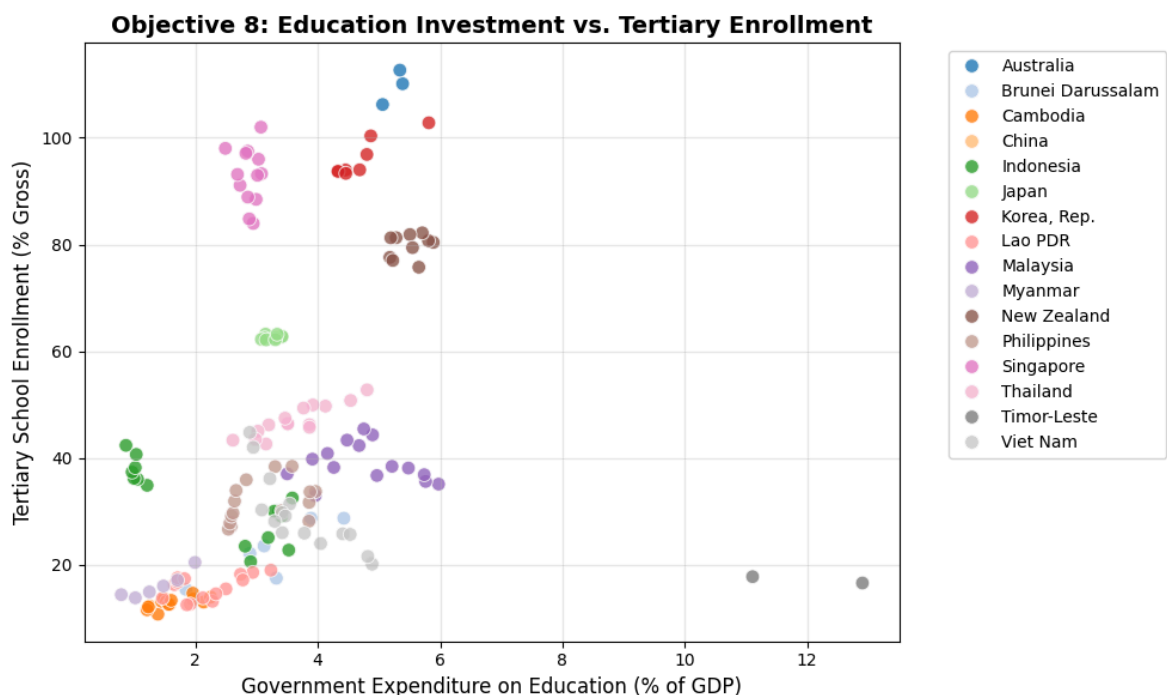
- **Chart Choice:** Scatter plot. Effectively maps the linear relationship between financial input (X-axis) and educational output (Y-axis) across numerical data

points to see if higher spending leads to higher enrollment.

```
In [14]: plt.figure(figsize=(10, 6))

sns.scatterplot(data=pivot_df,
                x='Government expenditure on education, total (% of GDP)',
                y='School enrollment, tertiary (% gross)',
                hue='Country Name',
                palette='tab20', s=70, alpha=0.8)

plt.title('Objective 8: Education Investment vs. Tertiary Enrollment', fo
plt.xlabel('Government Expenditure on Education (% of GDP)', fontsize=12)
plt.ylabel('Tertiary School Enrollment (% Gross)', fontsize=12)
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left', ncol=1)
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```



Analysis & Conclusion (Objective 8):

- **No Strict Correlation:** Higher spending does not automatically guarantee higher enrollment. For example, Timor-Leste spends the most (% of GDP) but has very low tertiary enrollment.
- **High Efficiency:** Countries like South Korea (Korea, Rep.) and Australia achieve massive tertiary enrollment (near or above 100%) with moderate expenditure (around 4-5.5% of GDP).
- **Low Input, Low Output:** Countries like Cambodia and Myanmar cluster at the bottom left, showing that very low government investment aligns with very low tertiary enrollment.

Objective 9: Monitor expansion of tertiary education access

- **Chart Choice:** Line chart. Ideal for comparing multi-year growth trajectories for time-series data across categorical entities (nations). We will highlight specific nations with notable trajectories.

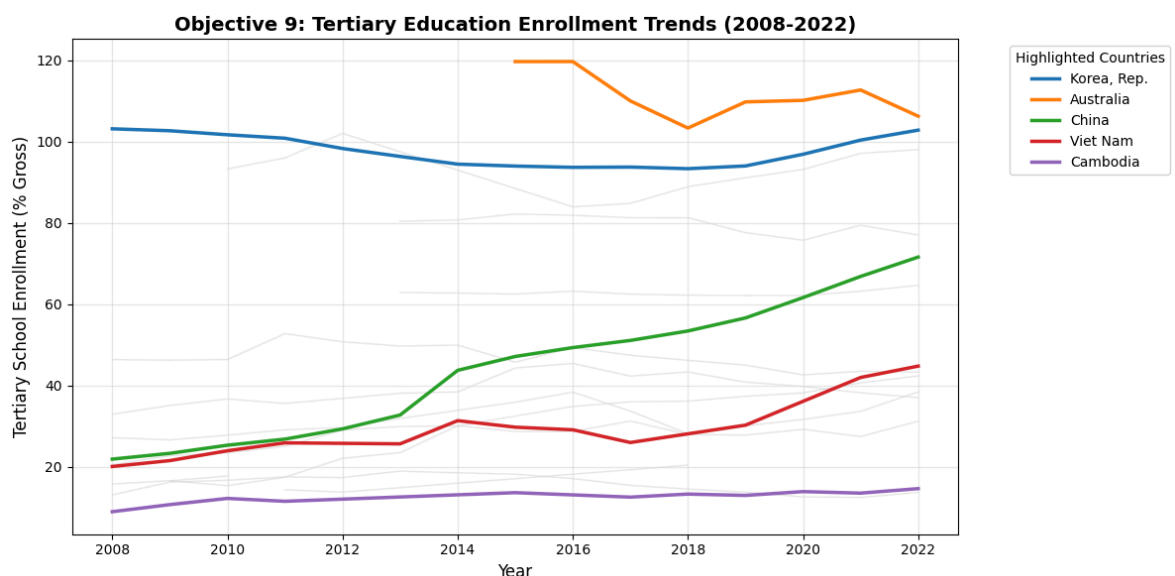
```
In [15]: plt.figure(figsize=(12, 6))

# Highlight countries with interesting trends
highlight_edu = ['Korea, Rep.', 'Australia', 'China', 'Viet Nam', 'Cambodia']
palette_edu = sns.color_palette("tab10", len(highlight_edu))

# Plot background
for country in pivot_df['Country Name'].unique():
    if country not in highlight_edu:
        country_data = pivot_df[pivot_df['Country Name'] == country]
        sns.lineplot(data=country_data, x='Year', y='School enrollment, t',
                      color='lightgray', alpha=0.5, linewidth=1)

# Plot highlights
for idx, country in enumerate(highlight_edu):
    country_data = pivot_df[pivot_df['Country Name'] == country]
    sns.lineplot(data=country_data, x='Year', y='School enrollment, terti',
                  color=palette_edu[idx], label=country, linewidth=2.5)

plt.title('Objective 9: Tertiary Education Enrollment Trends (2008–2022)')
plt.xlabel('Year', fontsize=12)
plt.ylabel('Tertiary School Enrollment (% Gross)', fontsize=12)
plt.legend(title='Highlighted Countries', bbox_to_anchor=(1.05, 1), loc='')
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```



Analysis & Conclusion (Objective 9):

- **General Upward Trend:** Almost all countries show an increase in tertiary enrollment over the 15 years.
- **Rapid Expansion:** China shows the most aggressive growth, shooting up from roughly 20% in 2008 to over 70% by 2022.
- **Top Performers:** Australia and South Korea remain at the very top, consistently hovering near or above 100% enrollment.

- **Slow Growth:** Viet Nam and Cambodia show relatively slow growth, reaching only roughly 45% and 15% respectively by the end of the period.

Theme 4: Infrastructure and Public Health

Objective 10: Measure progress in rural electrification

- **Chart Choice:** Line chart. Highlights infrastructural catch-up progress for continuous time-series data, focusing on developing nations while contrasting them with a developed nation (e.g., Japan) that has achieved and maintained full coverage.

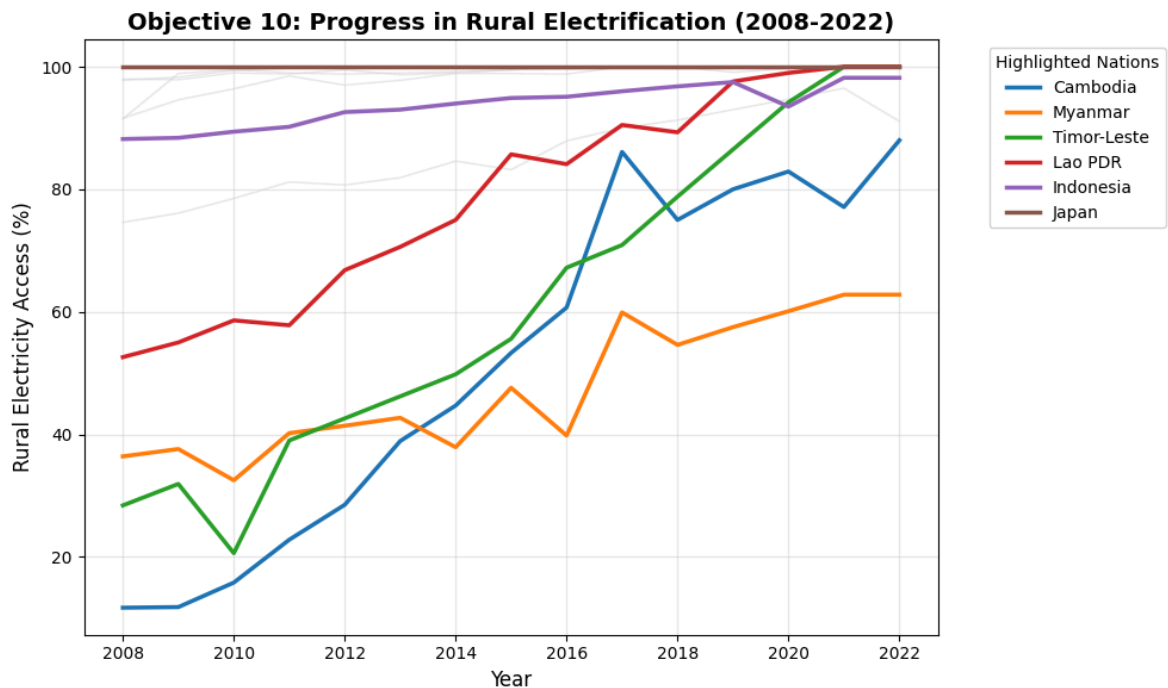
```
In [16]: plt.figure(figsize=(10, 6))

# Highlight developing nations showing progress, along with Japan as a ba
highlight_elec = ['Cambodia', 'Myanmar', 'Timor-Leste', 'Lao PDR', 'Indon
palette_elec = sns.color_palette("tab10", len(highlight_elec))

# Plot background
for country in pivot_df['Country Name'].unique():
    if country not in highlight_elec:
        country_data = pivot_df[pivot_df['Country Name'] == country]
        sns.lineplot(data=country_data, x='Year', y='Access to electricit
                        color='lightgray', alpha=0.5, linewidth=1)

# Plot highlights
for idx, country in enumerate(highlight_elec):
    country_data = pivot_df[pivot_df['Country Name'] == country]
    sns.lineplot(data=country_data, x='Year', y='Access to electricity, r
                  color=palette_elec[idx], label=country, linewidth=2.5)

plt.title('Objective 10: Progress in Rural Electrification (2008–2022)',
plt.xlabel('Year', fontsize=12)
plt.ylabel('Rural Electricity Access (%)', fontsize=12)
plt.legend(title='Highlighted Nations', bbox_to_anchor=(1.05, 1), loc='up
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```



Analysis & Conclusion (Objective 10):

- **The Developed Baseline (Japan):** Japan represents the infrastructure goal, maintaining a consistent 100% rural electricity access from 2008 onwards. The gray lines confirm most developed nations follow this identical flat baseline.
- **Massive Improvements:** Cambodia and Timor-Leste show incredible infrastructural growth. Cambodia grew from near 10% to nearly 90%, and Timor-Leste caught up rapidly from roughly 30% to a full 100%.
- **Catch-up phase:** Lao PDR and Indonesia also rapidly closed the infrastructure gap (reaching 100%). Meanwhile, despite progress, Cambodia and Myanmar remain the lowest in the region compared to the baseline.

Objective 11: Evaluate water access and life expectancy relationship

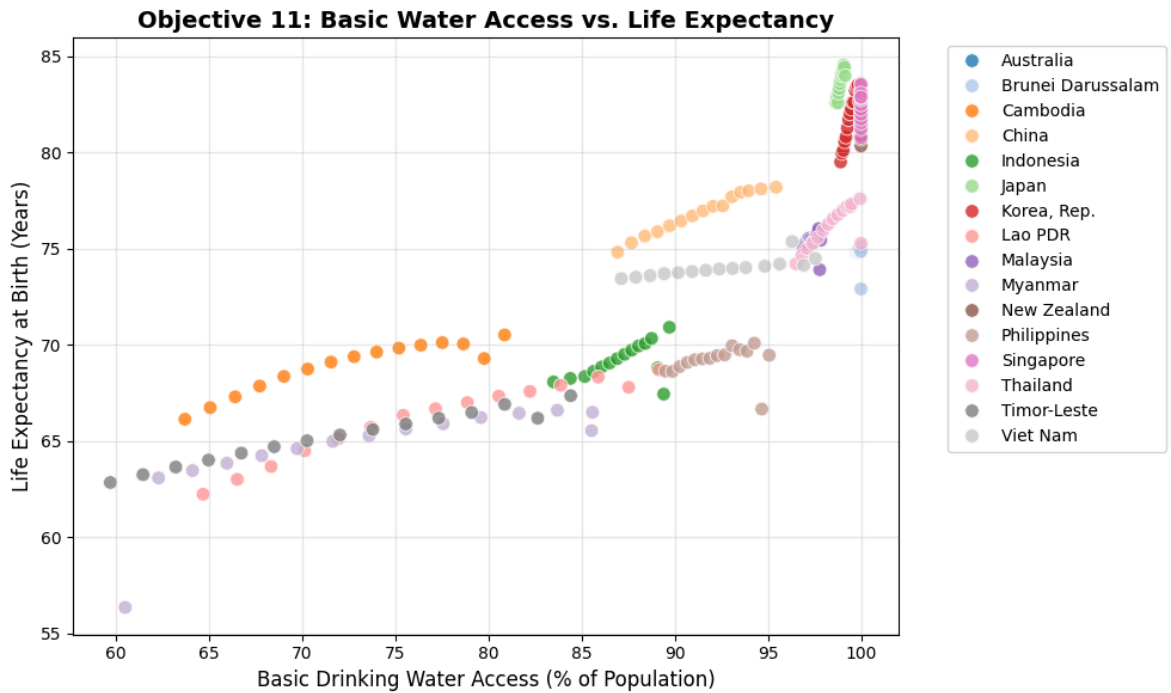
- **Chart Choice:** Scatter plot. Clearly reveals clusters and positive correlations between two continuous numerical variables in global public health data. This maps basic infrastructure (water) directly against human health outcomes (lifespan) to test for structural linkages.

```
In [17]: plt.figure(figsize=(10, 6))

sns.scatterplot(data=pivot_df,
                x='People using at least basic drinking water services (%)',
                y='Life expectancy at birth, total (years)',
                hue='Country Name',
                palette='tab20', s=70, alpha=0.8)

plt.title('Objective 11: Basic Water Access vs. Life Expectancy', fontsize=14)
plt.xlabel('Basic Drinking Water Access (% of Population)', fontsize=12)
plt.ylabel('Life Expectancy at Birth (Years)', fontsize=12)
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left', ncol=1)
plt.grid(True, alpha=0.3)
```

```
plt.tight_layout()
plt.show()
```



Analysis & Conclusion (Objective 11):

- **Strong Positive Cluster:** There is a clear, strong positive relationship. As drinking water access approaches 100%, life expectancy tightly clusters at higher ages (75 to 85 years).
- **Low Access, Low Lifespan:** Data points on the bottom left (mostly representing earlier years for countries like Timor-Leste, Myanmar, and Cambodia) show that lower water access (60-70%) strictly correlates with lower life expectancy (55-65 years). Notably, Myanmar consistently records the lowest outcomes on both metrics.

Objective 12: Compare the distribution of life expectancy across countries

- **Chart Choice:** Violin plot. Ideal for comparing the probability distribution and central tendency (median/quartiles) of numerical data across multiple categories (countries). The violin plot, showing density curves, clearly illustrates exactly where the lifespan data clusters for each nation.

```
In [18]: plt.figure(figsize=(10, 8))

# Calculate median life expectancy for sorting
median_life = pivot_df.groupby('Country Name')['Life expectancy at birth,

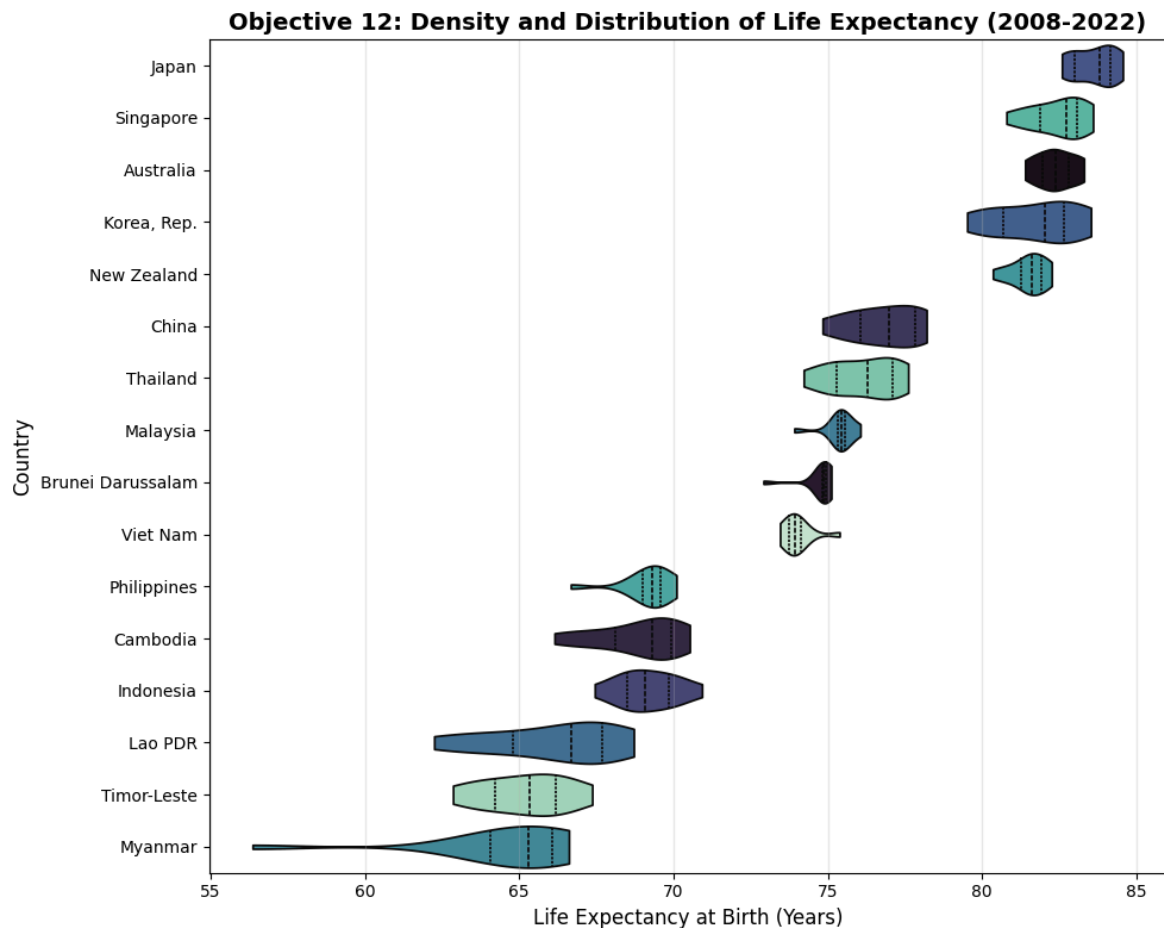
# Violin plot instead of Box plot
sns.violinplot(data=pivot_df,
               x='Life expectancy at birth, total (years)',
               y='Country Name',
               hue='Country Name',
               legend=False,
               order=median_life.index,
```

```

palette='mako', inner="quartile", cut=0)

plt.title('Objective 12: Density and Distribution of Life Expectancy (2008-2022)')
plt.xlabel('Life Expectancy at Birth (Years)', fontsize=12)
plt.ylabel('Country', fontsize=12)
plt.grid(axis='x', alpha=0.3)
plt.tight_layout()
plt.show()

```



Analysis & Conclusion (Objective 12):

- Strong Medical Disparity:** There is a significant, structural >20-year gap between top performers (Japan >83 years, Australia) and the vulnerable economies at the bottom (Myanmar and Timor-Leste around 64-67 years).
- Significant Improvements & Broad Distribution (Bottom nations):** While Myanmar, Timor-Leste, and Cambodia sit at the bottom, their density curves are exceptionally long and stretched, indicating substantial upward progress in health outcomes over the 15-year period. (Notably, Myanmar's violin has a severe downward tail dropping near 56 years, reflecting the devastating human toll of Cyclone Nargis in 2008).
- Structural Ceiling (Top nations):** Japan, Singapore, and Australia have "fat", short density bulges directly at the top. They have clearly hit a biological/medical ceiling around 83-84 years, exhibiting virtually zero variation and maintaining extreme stability over the entire period.

Theme 5: Multi-Dimensional Synthesis

Objective 13: Build a cross-indicator correlation overview

- **Chart Choice:** Correlation heatmap. Provides a global view of pairwise correlation coefficients across all continuous numerical variables in a single matrix. **Red** indicates positive correlations, while **blue** indicates negative correlations

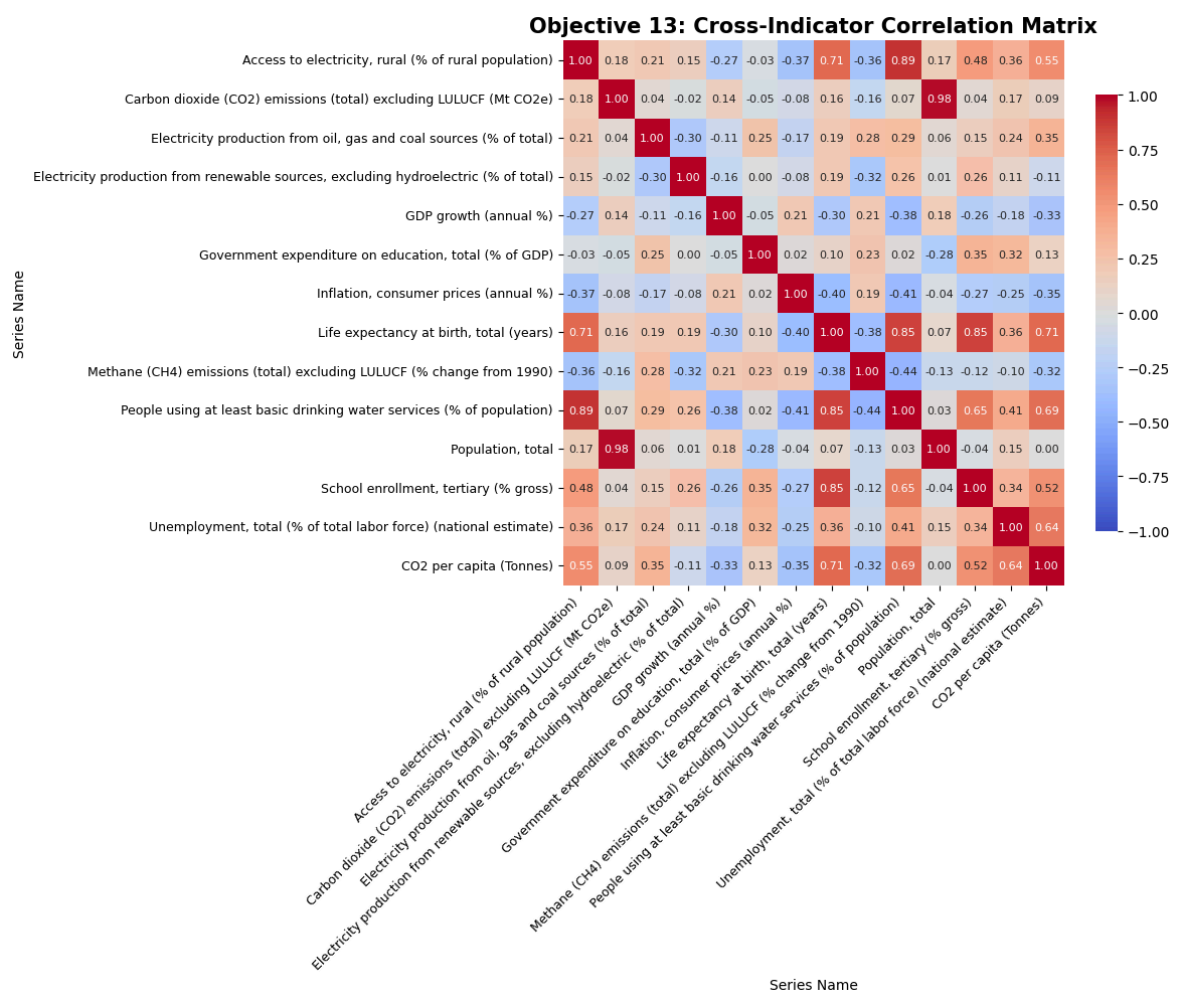
```
In [19]: plt.figure(figsize=(12, 10))

# Select only numeric columns for correlation (excluding Year to focus on
numeric_df = pivot_df.drop(columns=['Country Name', 'Year']).select_dtype

# Calculate correlation matrix
corr_matrix = numeric_df.corr()

# Create a clean heatmap
sns.heatmap(corr_matrix, annot=True, fmt=".2f", cmap='coolwarm', vmin=-1,
            annot_kws={"size": 8}, cbar_kws={'shrink': .8})

plt.title('Objective 13: Cross-Indicator Correlation Matrix', fontsize=15)
plt.xticks(rotation=45, ha='right', fontsize=9)
plt.yticks(fontsize=9)
plt.tight_layout()
plt.show()
```



Analysis & Conclusion (Objective 13):

- **Strongest Positive Links:** "Life expectancy" and "Basic drinking water access" have an extremely strong positive correlation (0.85), confirming the findings in Objective 11. "Tertiary school enrollment" also strongly correlates with both life expectancy (0.85) and water access (0.65).
- **Negative Links:** "Inflation" has a notable negative correlation with both "Life expectancy" (-0.40) and "Water access" (-0.41). Additionally, "GDP growth" shows a weak negative correlation with "Water access" (-0.38), "Life expectancy" (-0.3) and "CO2 per capita" (-0.33), suggesting that rapid economic growth periods may not immediately lead to improved environmental efficiency or basic infrastructure distribution.

Objective 14: Compare multi-dimensional country profiles

- **Chart Choice:** Radar chart (Spider chart). Perfect for synthesizing multi-dimensional numerical data into a profile to compare categorical entities (countries). We will map out 5 core development pillars for 4 representative nations.
- **Design Choice:** To ensure that indicators with massively different units do not geometrically distort the shape, we must first collapse the measurements into country averages across the 15 years, and apply **Min-Max Scaling (0 to 1)** to normalize the axes. A perfectly balanced development footprint will resemble a large, uniform polygon.

```
In [20]: from sklearn.preprocessing import MinMaxScaler

# Select metrics for multi-dimensional profile
radar_metrics = [
    'GDP growth (annual %)',
    'CO2 per capita (Tonnes)',
    'School enrollment, tertiary (% gross)',
    'Access to electricity, rural (% of rural population)',
    'Life expectancy at birth, total (years)'
]

# Calculate averages for each country
avg_df = pivot_df.groupby('Country Name', as_index=False)[radar_metrics].
avg_df = avg_df.dropna() # Just to be safe if any country dropped everyth

# 1. Min-Max Scaling (0 to 1) for the Radar Chart
scaler = MinMaxScaler()
scaled_data = scaler.fit_transform(avg_df.iloc[:, 1:])

# Convert back to DataFrame
scaled_df = pd.DataFrame(scaled_data, columns=radar_metrics)

# FIX: Invert CO2 metric so further from center (1.0) means HIGH efficiency
scaled_df['CO2 per capita (Tonnes)'] = 1 - scaled_df['CO2 per capita (Tonnes)']

scaled_df['Country Name'] = avg_df['Country Name'].values

# Define countries to contrast (e.g. Developed vs Emerging vs Vulnerable)
highlight_radar = ['Australia', 'Japan', 'Viet Nam', 'Timor-Leste']
```

```

# 2. Plotting the Radar Chart
fig, ax = plt.subplots(figsize=(8, 8), subplot_kw=dict(polar=True))

# Create an angle for each metric axis
num_vars = len(radar_metrics)
angles = np.linspace(0, 2 * np.pi, num_vars, endpoint=False).tolist()
angles += angles[:1] # Complete the circle by appending the first angle

# Clean up metric names for the chart labels (Changed to CO2 Efficiency)
short_labels = ['GDP Growth', 'CO2\nEfficiency', 'Tertiary Edu\n(% Enroll

colors = sns.color_palette("tab10", len(highlight_radar))

for idx, country in enumerate(highlight_radar):
    values = scaled_df[scaled_df['Country Name'] == country].iloc[0, :-1]
    values += values[:1] # Close the polygon

    ax.plot(angles, values, color=colors[idx], linewidth=2, label=country)
    ax.fill(angles, values, color=colors[idx], alpha=0.1)

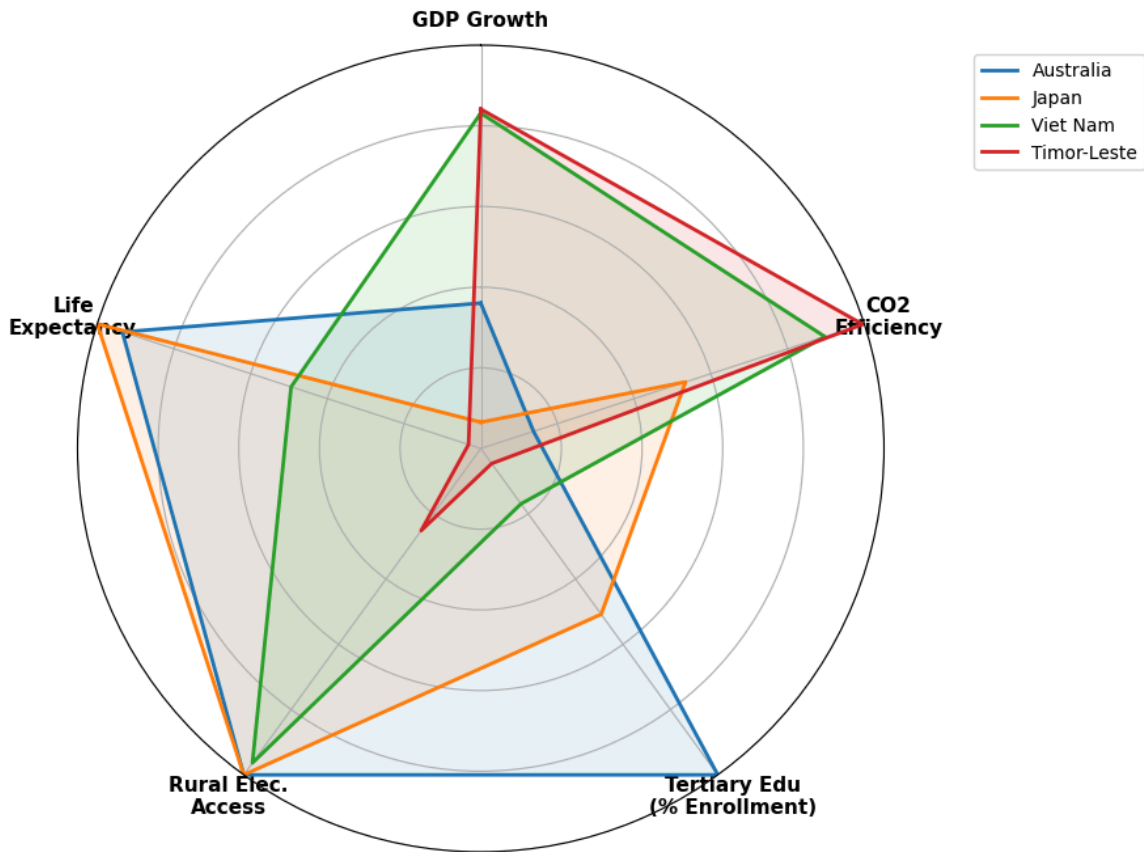
# Format the grid and labels
ax.set_theta_offset(np.pi / 2) # Top is 0 degrees
ax.set_theta_direction(-1) # Clockwise
ax.set_thetagrids(np.degrees(angles[:-1]), short_labels, fontsize=11, fon

ax.set_ylim(0, 1) # Normalised strictly between 0 and 1
ax.set_rgrids([0.2, 0.4, 0.6, 0.8, 1], angle=0, labels=[]) # Remove gener

plt.title('Objective 14: Multi-Dimensional Development Profiles (Normaliz
plt.legend(bbox_to_anchor=(1.1, 1), loc='upper left')
plt.show()

```

Objective 14: Multi-Dimensional Development Profiles (Normalized)



Analysis & Conclusion (Objective 14):

- **High Development, Low Efficiency (Australia):** Australia scores highly on Tertiary Education, Life Expectancy, and Rural Electrification. However, its polygon shrinks severely on CO2 Efficiency, highlighting a high per-capita environmental cost for its development.
- **Developed but Stagnant (Japan):** Japan exhibits a balanced, highly developed profile (maximized life expectancy and electrification, moderate CO2 efficiency) but shows significant weakness in GDP growth, reflecting economic stagnation.
- **Rapid Emerging Catch-up (Viet Nam):** Viet Nam displays strong GDP growth, high CO2 efficiency, and maximizes rural electrification. Its primary developmental lag remains in Tertiary Education, which forms the smallest point on its polygon.
- **Early Development Stage:** Timor-Leste has the most contracted overall profile. While it maximizes CO2 efficiency (due to negligible absolute emissions) and registers the highest peak volatility in GDP growth, it lags significantly behind the others in Education, Life Expectancy, and Rural Electrification compared to Viet Nam, with similar GDP growth, ..etc but but much higher Edu, Life and Electric value.

Objective 15: Explore the modernization trade-off (Quality of Life vs. Emissions)

- **Chart Choice:** Scatter plot. Maps Life Expectancy against CO2 per capita to analyze the relationship between per-capita environmental impact and human health outcomes.

```
In [21]: plt.figure(figsize=(10, 6))

if 'CO2 per capita (Tonnes)' not in pivot_df.columns:
    pivot_df['CO2 per capita (Tonnes)'] = (pivot_df['Carbon dioxide (CO2)

# Use 2022 subset for a clean Scatter chart
scatter_df = pivot_df[pivot_df['Year'] == 2022].copy()

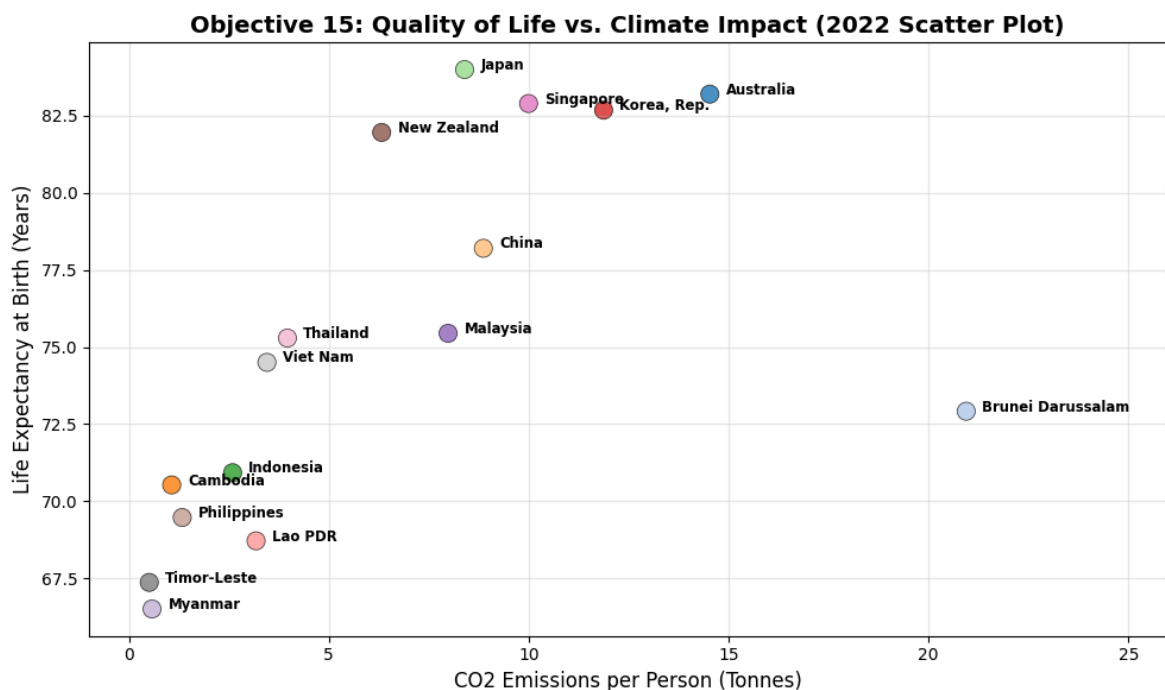
sns.scatterplot(data=scatter_df,
                x='CO2 per capita (Tonnes)',
                y='Life expectancy at birth, total (years)',
                hue='Country Name',
                s=120, # Fixed dot size
                palette='tab20', alpha=0.8,
                legend=False, edgecolor='black', linewidth=0.5)

# Adding annotations for readability
for i in range(scatter_df.shape[0]):
    text_x = scatter_df['CO2 per capita (Tonnes)'].iloc[i] + 0.4
    text_y = scatter_df['Life expectancy at birth, total (years)'].iloc[i]
    if pd.notnull(text_x) and pd.notnull(text_y):
        plt.text(text_x, text_y,
                 scatter_df['Country Name'].iloc[i],
                 horizontalalignment='left', size='small', color='black',

plt.title('Objective 15: Quality of Life vs. Climate Impact (2022 Scatter
plt.xlabel('CO2 Emissions per Person (Tonnes)', fontsize=12)
plt.ylabel('Life Expectancy at Birth (Years)', fontsize=12)

x_max = scatter_df['CO2 per capita (Tonnes)'].max() + 5
plt.xlim(left=-1, right=x_max)

plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```



Analysis & Conclusion (Objective 15):

- **Health-Emission Trade-off:** A broad positive correlation exists between carbon footprint and health outcomes. Nations with lower life expectancies (e.g., Cambodia, Lao PDR, Timor-Leste) cluster tightly on the left with very low per-capita CO2 emissions (<5 tonnes). As life expectancy pushes beyond 75 years, emissions per person generally stretch further right.
- **High-End Inefficiency (Australia & Brunei):** Environmental efficiency varies widely across modern nations. Japan achieves the peak ~84-year life expectancy efficiently at roughly 8.5 tonnes of CO2 per capita, and Singapore does similarly well (near 83 years) at 10.0 tonnes. In contrast, Australia generates nearly 14.5 tonnes per capita for a similar high lifespan. Most strikingly, Brunei produces 21 tonnes per capita while achieving a significantly lower health outcome (~73 years), heavily separated from the main cluster and highlighting a highly inefficient model of modernization.

4. Team Contributions

Name	Task	Contribution
Tâm	Objectives 7-9 Data cleaning & preprocessing	100%
Lam	Objectives 13-15 Video taping Data cleaning & preprocessing	100%
Chi	Objectives 4-6 Notebook Structuring	100%
Thắng	Objectives 1-3 Video taping	100%
Thượng	Objectives 10-12 Determine the dataset scope	100%

5. References & Acknowledgments

5.1. References

- **Data Source:** World Bank Group (2024). *World Development Indicators (WDI)* - Asia Pacific territory (2008-2022 dataset).
- **Libraries:** `pandas` (McKinney, 2010), `matplotlib` (Hunter, 2007), `seaborn` (Waskom, 2021), `scikit-learn` (Pedregosa, 2011).

5.2. Acknowledgments

- **Instructor:** Sincere thanks to our instructor, Mr. **Vo Nhat Tan**, for guiding us through the Data Visualization course and providing valuable insights that enhanced our understanding.

- **AI Assistance:** This project's code structuring, data exploration, and debugging processes were greatly assisted by AI chatbots (**Google Gemini** via GitHub Copilot).